Geography Ch-4 — Weathering, Mass Movement & Groundwater

Physical Geography | Denudation Processes | GS-I

FOUNDATION: G.C. Leong · STANDARD: Majid Husain (Indian & World Geography)

Register-style revision notes · grounded in source books

Facts from official sources | Inferences clearly marked | No fabricated data

■ HIGH UPSC RELEVANCE

HIGH

FOUNDATION — Weathering, Mass Movement & Groundwater (G.C. Leong)

GS-I · Physical Geography → Denudation (Source: G.C. Leong, Certificate Physical & Human Geography)

■ NEWS TRIGGER

Dharali (Uttarkashi, Uttarakhand) debris-flow disaster, 5 Aug 2025 — a rainfall-triggered mudslide destroyed ~148 buildings; and the Aug 2025 Katra landslide near Vaishno Devi (~30 dead) put slope failure / mass movement in the news.

■ INTRO & ORIGIN

Weathering is the **in-situ** disintegration and decomposition of rock at or near the Earth's surface — without transport. It differs from **erosion**, which involves the removal and transport of the loosened debris by rivers, wind, ice or waves.

*Weathering prepares the loose mantle (regolith) that gravity and running water then move. Leong groups it into three families: **mechanical**, **chemical** and **biological**.*

■ KEY DATA

Type	Chief Agent / Process	Result (example)
Mechanical (physical)	Temperature change, frost action, unloading	Exfoliation ('onion weathering'); screes / scree slopes
Chemical	Solution, esp. rain-water + CO2 (weak acid)	Limestone dissolved along joints; caves widened
Biological	Plant roots in joints; burrowing / trampling animals	Rock split along bedding planes
Mass movement	Gravity on weathered debris (± rain lubrication)	Soil creep (slow); landslides/slumping (rapid)
Groundwater	Seepage to impermeable layer	Water-table, aquifer, springs, artesian wells

■ STATIC THEORY

• **Exfoliation:** outer rock heats and peels off in thin layers like an onion; angular debris collects as **scree** at cliff foot.

• **Frost action:** water freezing in cracks expands ~10% of its volume and prises off fragments (important in cold/high-altitude climates).

• **Mass movement** = down-slope movement of weathered material under gravity; speed depends on slope gradient, debris weight and rain lubrication.

• **Soil creep:** slow, near-continuous soil movement — tilts trees, fences, posts; debris piles at slope foot.

• **Landslides (slumping/sliding):** rapid fall of a large soil/rock mass on steep or undercut slopes; triggered by rivers/sea undercutting, earthquakes, or man-made cuttings.

• **Groundwater:** water seeps down until an impermeable layer stops it; the saturated permeable rock storing it is the **aquifer**, its upper surface the **water-table**.

• **Springs** = natural outlets where the water-table meets the surface; an **artesian well** forms where an aquifer lies between two impermeable strata and water rises under pressure.

■ **MUST-KNOW FACTS**

1. Weathering is IN-SITU (no transport); erosion = weathering + transport.

2. Exfoliation = 'onion weathering'; debris = scree.

3. Frost water expands ~10% by volume → freeze-thaw shattering.

4. Solution is the most potent process in LIMESTONE regions.

5. Aquifer = water-bearing permeable rock; water-table = its saturated surface.

6. Artesian well: aquifer sandwiched between two impermeable layers → water rises without pumping.

■■ **UPSC TRAPS**

✗ Weathering and erosion are the same.

✓ Weathering is in-situ breakdown; erosion adds transport of debris.

✗ Soil creep is a rapid, visible movement.

✓ Soil creep is SLOW/imperceptible; landslides are the rapid form of mass movement.

✗ All wells are artesian.

✓ Only where a confined aquifer lies between two impermeable strata does water rise under natural pressure (artesian).

■ **MAINS ANGLE**

Understanding weathering + mass movement explains why fragile, over-steepened Himalayan slopes (Dharali, Katra) fail during intense monsoon rain — a base for disaster-vulnerability answers.

■ **STUDY LINK**

Geography → Physical → Denudation → Weathering & Mass-wasting (Leong Ch-4)

HIGH

STANDARD — Weathering Controls, Chemical Processes & Karst Hydrology (Majid Husain)

GS-I · Geomorphology → Denudation (Source: Majid Husain, Indian & World Geography)

■ **NEWS TRIGGER**

Scientists link the rising frequency/severity of 2025 Himalayan mass-movement events to climate change intensifying rainfall — raising demand for hazard zonation of weathering-weakened slopes.

■ **INTRO & ORIGIN**

Rate and depth of weathering are controlled by six factors (Majid Husain): **(i) rock structure & hardness, (ii) climate, (iii) topography & slope, (iv) natural vegetation, (v) time, and (vi) land use by man.**

Climate is decisive: ***physical/mechanical*** *weathering dominates high latitudes & altitudes (frost, insolation), while* ***chemical*** *weathering dominates hot-humid equatorial, monsoon and sub-tropical wet regions — and is generally the more important of the two.*

■ **KEY DATA**

Chemical process	Mechanism	Diagnostic outcome
Solution	Limestone (CaCO3) passes into liquid state with water	Widened joints/grikes; karst
Carbonation	Weak carbonic acid (H2O + CO2) attacks Ca/Mg/K/Na minerals	Carbonates formed; limestone decay
Hydrolysis	Minerals chemically combine with water	Feldspar → clay minerals
Oxidation	O2 (via water) reacts with minerals (e.g. iron)	Reddish/rusty rotted rock

■ **STATIC THEORY**

• **Frost action** works where water expands ~9% on freezing — powerful in humid micro-thermal, sub-arctic, polar and high-altitude climates.

• **Crystallisation** (salt weathering) and **exfoliation domes** (concentric shells stripped from a rock mass) are advanced physical forms.

• **Chemical weathering is more pronounced in karst / limestone regions** because solution + carbonation are highly effective there.

• **Karst** is typified by dominant solution, LACK of surface water, and stream sinks (dolines), cave systems and resurgences (springs); it lacks a well-integrated surface drainage.

• Karst development needs thick, well-jointed, pure limestone at/above the water-table in a humid climate (conditions for maximum solution).

• Groundwater zones: zone of aeration above, **zone of saturation** (permeable rock) below the water-table; springs emerge where the table is cut by the surface (scarp-foot, dip-slope, dyke, Vauclusian springs).

■ MUST-KNOW FACTS

1. Six weathering controls: rock structure, climate, slope, vegetation, time, land use (Majid Husain).

2. Chemical weathering > physical in hot-humid (equatorial/monsoon) climates.

3. Four chemical processes: SOLUTION, CARBONATION, HYDROLYSIS, OXIDATION.

4. Carbonation uses weak carbonic acid (water + CO_2) on limestone.

5. Karst signatures: dolines (stream sinks), caves, resurgences; poor surface drainage.

6. Frost action peaks in micro-thermal / sub-arctic / polar / high-altitude zones.

■■ UPSC TRAPS

✗ Carbonation and solution are identical.

✓ Solution = limestone dissolving into liquid; carbonation = carbonic acid reacting to form carbonates — related but distinct.

✗ Karst regions have dense surface drainage.

✓ Karst LACKS integrated surface drainage — water sinks underground via dolines to form caves and resurgences.

✗ Chemical weathering is strongest in cold dry climates.

✓ It is strongest in hot-humid climates; cold/high-altitude zones favour physical (frost) weathering.

■ MAINS ANGLE

Contrast the climatic control of weathering (physical in cold/high altitude vs chemical in hot-humid) and analyse karst hydrology's paradox — water-rich rock yet surface water scarcity — for resource & hazard answers.

■ STUDY LINK

Geography → Geomorphology → Denudation processes (Majid Husain)

End of Register Notes — UPSC Agent / Copilot CLI